# Testing on Thin Ice

## Chipping Away at Test Unpredictability

**François Martin**

# $36 Billion/year (USD)

Loss Productivity

# $40 Billion/year (USD)

End World Hunger
by 2030

# Topics

What is a Flaky Test?

How to find them

How to deal with them

How to fix them



Slides & eBook

# What is a Flaky Test?

No Flake News!

François Martin




# Definition

Test that exhibits both a passing and failing test result with the same code if run multiple times



# Timing related

Example:

1. Enter search term into textbox
2. Click on button “Search”
3. Wait for one second
4. Click on first result



# Test data related

Examples:

- Input field for prename doesn't allow special characters
  - Some randomly generated prenames in tests contain special characters ("François")
- Tests reliant on predefined test data in the database
  - Dependencies between tests
  - Failing tests cause inconsistencies



# Infrastructure related

Unstable infrastructure

Tests time out

Most difficult to reproduce

Examples:

- Build runners under heavy load
- Microservices with networking issues



# Resource contention related

Mostly when running tests in parallel

Examples: Multiple tests trying to...

- ... write to the same file at the same time
  - File lock
  - Concurrent modification errors
- ... create a user with the same username at the same time
  - Unique constraint violations



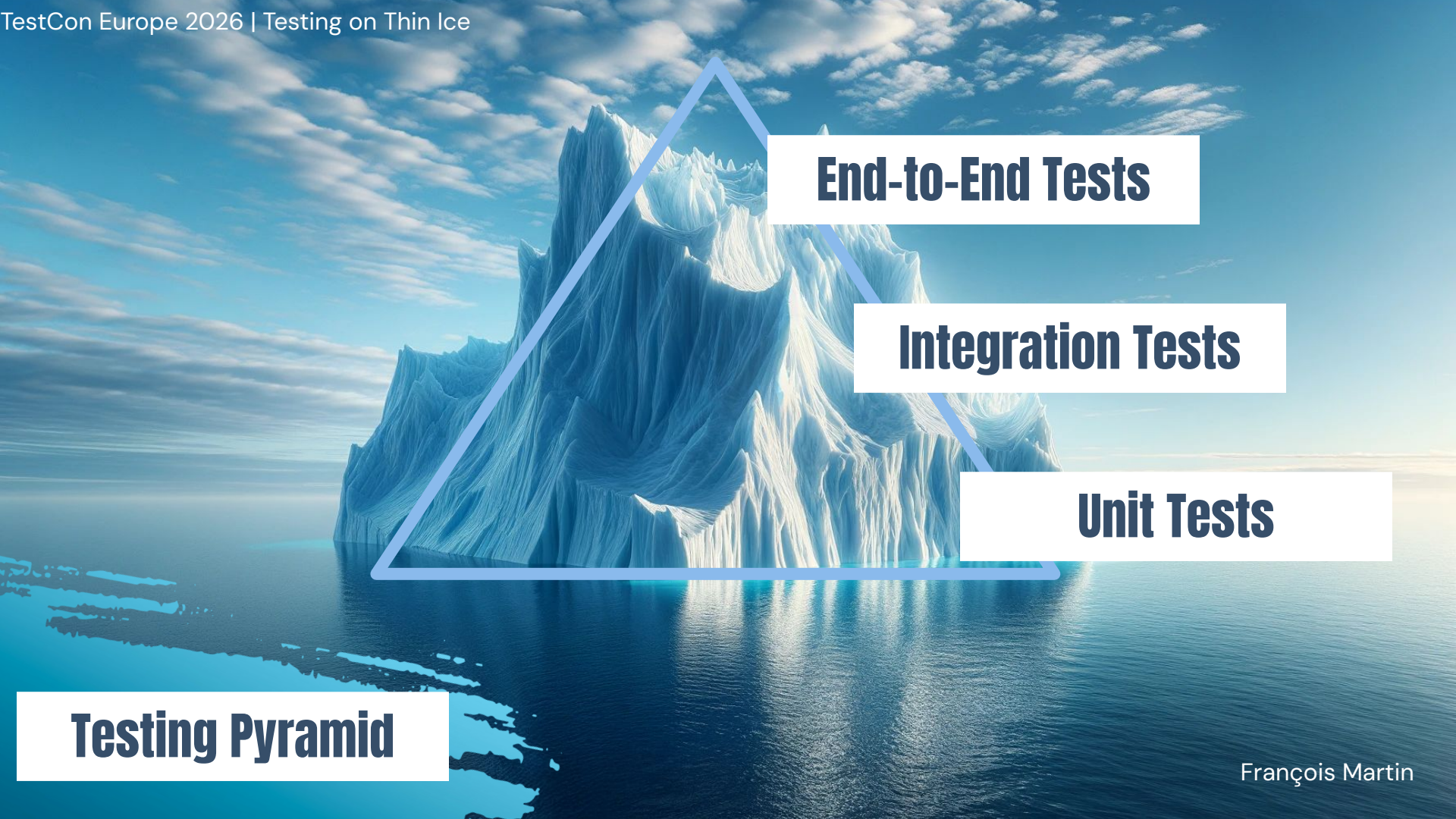

# Why Worry about Flaky Tests?

- Re-running tests wastes a lot of time
- Repetitive investigation efforts to determine if flaky or legitimate failure
- Team starts losing trust in test results
- Bad software quality
- Pilots frequently ignoring an alarm in cockpits of a Boeing 737 resulted in fatal plane crash in 2005

# Unit Tests vs Integration Tests

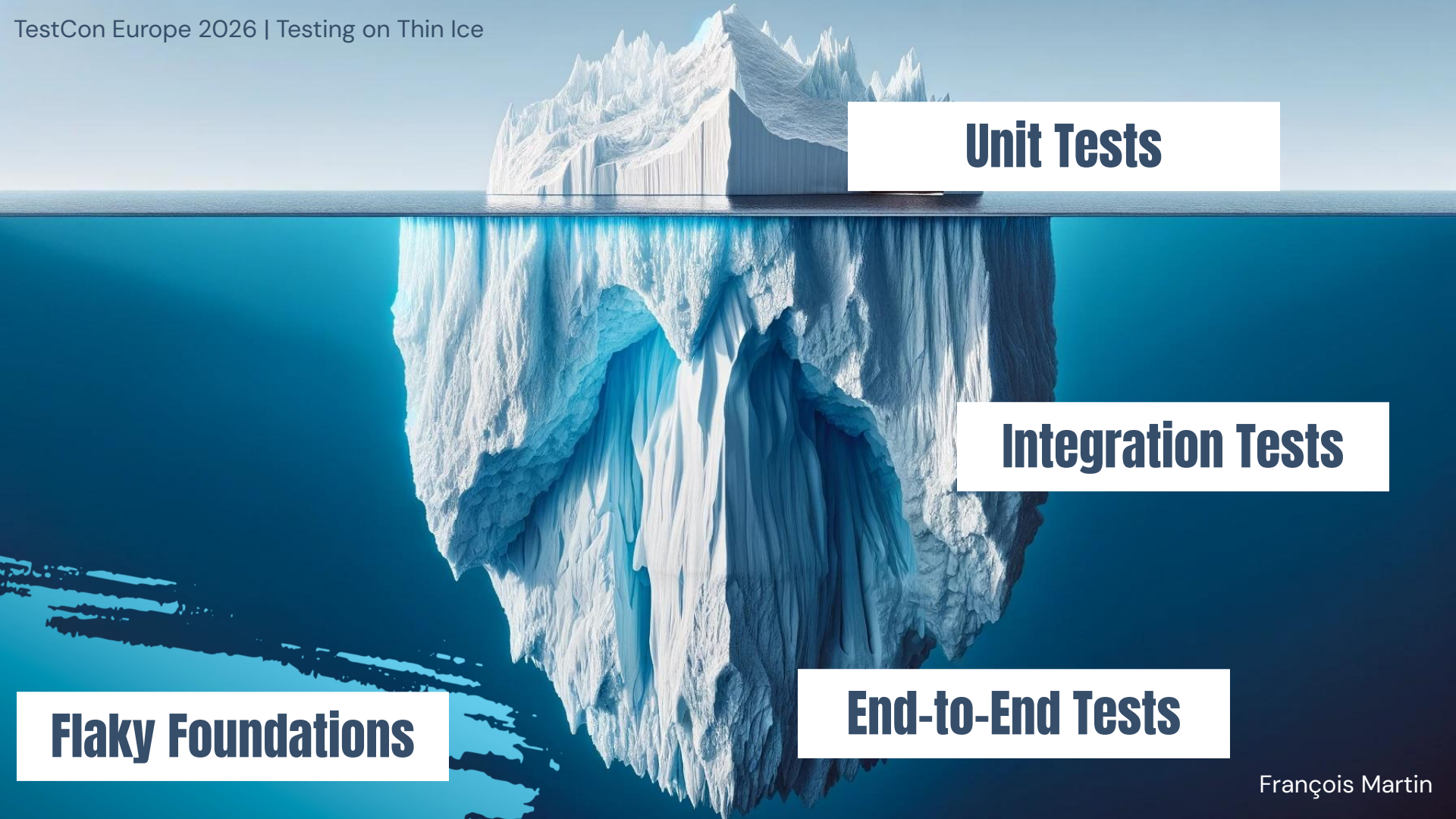
UT vs IT

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# Testing Pyramid

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Unit Tests
Integration Tests
End-to-End Tests
Flaky Foundations


Complexity
Unit Tests
Integration Tests
End-to-End Tests
Flakiness

# How to find Flaky Tests

Flake Busters

François Martin



# Running tests in parallel

Can reveal many issues:

- Resource contention
- Dependency on test data
- Dependency on test execution order

Side benefit: Tests execute faster 🚀

- Playwright & WebdriverIO: parallel by default
- Cypress: only in cloud (paid)
- JUnit and Gradle: through configuration



# Rerun Tests

Most test frameworks allow for executing a test multiple times.

## Slides & eBook

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# Maven Surefire Plugin

```
<plugin>
   <groupId>org.apache.maven.plugins</groupId>
   <artifactId>maven-surefire-plugin</artifactId>
   <configuration>
      <rerunFailingTestsCount>3</rerunFailingTestsCount>
      <failOnFlakeCount>0</failOnFlakeCount>
    </configuration>
</plugin>
```



# Rerun Gotcha

Increased time to fail

François Martin



# Dealing with Flaky Tests

François Martin



# Prevention

In Pull / Merge Requests, run new or changed tests 3, 10, 50, or 1000 times (depending on resources and test type)

Alternatively run all tests multiple times nightly



# Tell Your Team!

Don't let them waste time

Uncover deeper issues

Saves morale


New Sprint Checklist
Fix a flaky test
Pull a new issue




# Flaky Test Days

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# Temporarily Quarantine Tests

```
// JUnit
@Disabled("quarantine: reason or link to issue here")
@Test
void flakyTest() {
}

// Jest
// quarantine: reason or link to issue here
it.skip('should throw an error', () => {
  expect(response).toThrowError(expected_error);
});
```




# Quarantine

- Quarantine test after first failure
- Fix the quarantined tests timely
- Set a limit for tests in quarantine
- Define time limit for quarantined tests



# Delete Flaky Tests

Does the test provide value?

Can it be fixed?

Keeping a flaky test provides negative value

# SSH into CI Agent

```
$ ssh ci-agent

cd /your-project

# rerun build
./mvnw clean install
```

# How to fix Flaky Tests

Defeating Murphy's Law

François Martin

# Test Data Consistency

```java
// JUnit 5

@BeforeEach
void setup() {
  // Create test data
}

@AfterEach
void teardown() {
  // Delete test data
}
```

```javascript
// Jest

beforeEach(() => {
  // Create test data
});

afterEach(() => {
  // Delete test data
});
```

# Test Data Consistency

```
// Using Testcontainers

// build.gradle
testImplementation "org.testcontainers:postgres:1.19.8"
testImplementation "org.testcontainers:junit-jupiter:1.19.8"

// JUnit Test
@Testcontainers
public class IntegrationTest {
    @Container
    PostgreSQLContainer<?> postgresContainer =
        new PostgreSQLContainer<>("postgres:16.3");
}
```



# Isolate Tests

# Don't Reuse System Resources

- Setting System Properties

```
System.setProperty("key", "value");
```

- Use unique files for each test

  temp directories, output files, etc.

- Use a random **free** ephemeral port (49152 – 65535)

```
new ServerSocket(0).getLocalPort()
```

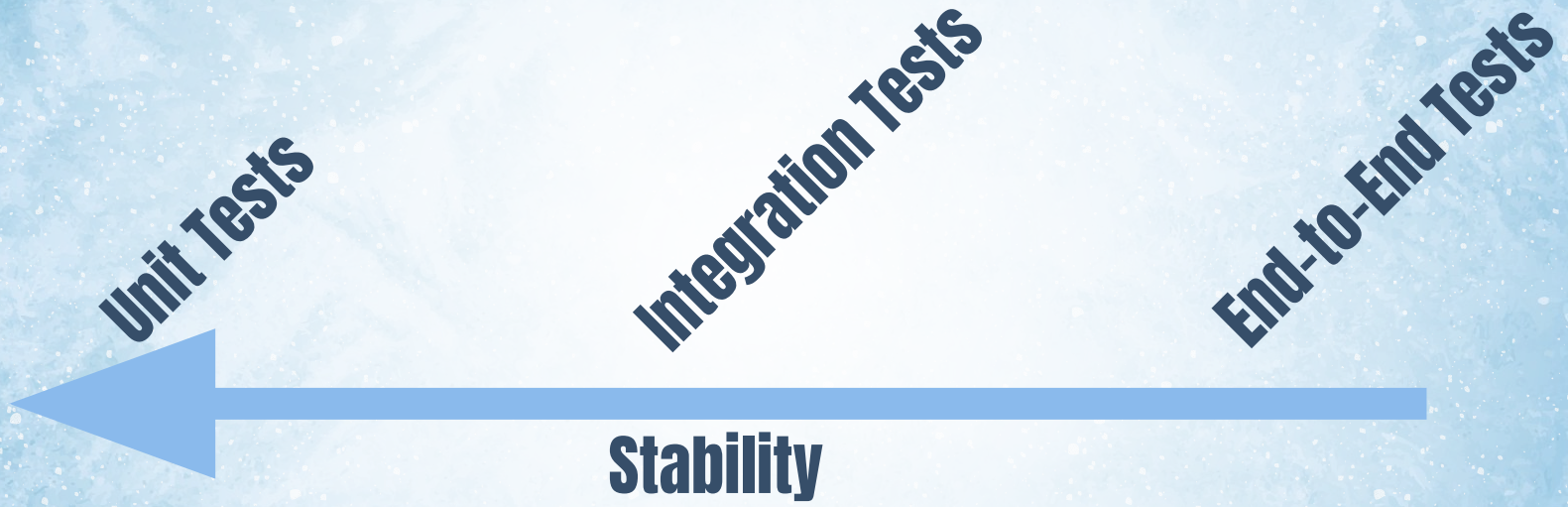
# In E2E Tests

- Use deep links to navigate directly to the page to be tested
- Start the test with the test user already logged in
- Make it possible to run each test spec independently
- Do not wait in predefined time intervals, wait for a condition to be fulfilled with a timeout
- Use stable test ids in selectors
- Consider mocking external APIs

# Convert ITs and E2E into UTs



# Convert ITs and E2E into UTs





# Fix the Culture of Flaky Tests

- Brown Bag / Lunch & Learn
- Make flaky tests visible
- Track them
- Use manager terms 💰

# QUESTIONS?

## KONTAKTAI

Slides & eBook

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